# Project Initialization and Planning Phase

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| **Date** | 23 October 2026 |
| **Team ID** | [Your Team ID] |
| **Project Title** | TravelMate - Personalized Travel Itinerary Planner |
| **Maximum Marks** | 5 Marks |

## Project Proposal (Proposed Solution) report

The proposal report aims to simplify trip planning using generative AI, giving travellers one place for a personalised itinerary, map, budget and PDF export. It tackles scattered planning tools, promising faster planning, better budget control and less stress. Key features include AI-generated day-wise itineraries, an interactive map and a budget tracker.

## Project Overview

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| **Objective** | To build a full-stack web application that generates a personalised, day-wise travel itinerary from the user's destination, dates, budget and preferences, and helps the user manage the trip. |
| **Scope** | User sign-in, trip creation and management, AI itinerary generation, itinerary viewer, interactive map with directions links, budget tracking and PDF export. |

## Problem Statement

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| **Description** | Travellers must use many websites to plan activities, routes and costs. This is slow, not personalised and makes budgets hard to control. |
| **Impact** | Solving this saves planning time, keeps spending visible, and gives travellers a confident, ready-to-use plan. |

## Proposed Solution

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| **Approach** | Use a Groq-powered AI service behind a FastAPI backend to turn trip preferences into a structured itinerary, store trips in Supabase, and present results in a React interface with a map, budget tracker and PDF export. |
| **Key Features** | - AI-generated day-wise itinerary with activities, timings, categories and costs.<br>- Trips dashboard with search, sort, edit and delete.<br>- Interactive Leaflet map with numbered pins and Google Maps directions.<br>- Budget tracker with remaining budget and daily average.<br>- Itinerary export to PDF. |

## Resource Requirements

| Resource Type | Description | Specification/Allocation |
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| **Hardware** | | |

| Resource Type | Description | Specification/Allocation |
|---|---|---|
| Computing Resources | CPU/GPU specifications, number of cores | Standard developer laptop; no local GPU (AI runs on Groq cloud) |
| Memory | RAM specifications | 8 GB |
| Storage | Disk space for code, dependencies and logs | 256 GB SSD |
| **Software** | | |
| Frameworks | Frontend and backend frameworks | React + Vite, FastAPI (Uvicorn) |
| Libraries | Additional libraries | Leaflet, jsPDF, Supabase client |
| Development Environment | IDE and tools | Antigravity IDE, Node.js, Python, Git |
| **Data** | | |
| Data | Source, size, format | AI-generated itineraries (JSON); trip data in Supabase; OpenStreetMap tiles; Nominatim geocoding |